Xingyu Zhou

Phone: +44 7918737368
E-mail: xz419@cam.ac.uk
Affiliation:
Department of Applied Mathematics and Theoretical Physics, University of Cambridge
Trinity College, University of Cambridge

Research interests

Direct numerical simulations (DNS) of stratified turbulence and shear-flow instability.
Lagrangian perspective of turbulent mixing.

Education Experience

2023-2027	PhD in Applied Mathematics and Theoretical Physics, University of Cambridge.
2022-2023	MMath (Honours Pass with Distinction), Mathematics Tripos Part III, Trinity College, Cambridge.
2019-2022	BA (First Class Honours), Mathematics Tripos Part IA - Part II, Trinity College, Cambridge.
2017-2019	Harrow School.

Publications

[1] Zhou X, Taylor JR, Caulfield CP. A Lagrangian view of mixing in stratified shear flows. Journal of Fluid Mechanics. 2025;1021:A31. doi:10.1017/jfm.2025.10626
[2] Zhou X, Taylor JR, Caulfield CP. Lagrangian inference of turbulent diffusivity in stratified shear flows. (Under revision in Journal of Fluid Mechanics.)

Conference

Ocean Sciences Meeting, Glasgow, UK, February 2026. *Poster*.
EGU General Assembly, Vienna, Austria, April 2025. *Oral presentation*.
APS Division of Fluid Dynamics, Salt Lake City, US, November, 2024. *Oral presentation, presented by Prof. Colm-cille Caulfield*.
CCfCS Winter Symposium, Cambridge, UK, November, 2024. *Poster*.

Awards and Scholarships

Trinity College Internal Graduate Scholarship, 2023-2027
Honorary Trinity-Henry Barlow Scholarship, 2023-24
Trinity College Senior Scholarship, 2023-2024

Trinity College Senior Scholarship, 2022-2023
Trinity College Senior Scholarship, 2021-2022
Gold Certificate, British Physics Olympiad Round 1, 2018
Gold Certificate, British Physics Olympiad Round 2, 2018
Gold Certificate, British Chemistry Olympiad Round 1, 2018
Silver Medal, British Mathematical Olympiad Round 1, 2018

Teaching and supervision

University of Cambridge, 2025-2026.
Supervisor for undergraduate students in Vectors and Matrices.
Part III Drop-In Session Leader in Continuum Mechanics.
University of Cambridge, 2024-2025.
Demonstrator for computational projects in The Fluid Dynamics of Sustainability and the Environment (FDSE) summer school.
Supervisor for undergraduate students in Vectors and Matrices.
Supervisor for undergraduate students in Vector Calculus.
Computer-Aided Teaching of All Mathematics (CATAM) Feedback Supervisor.
Part III Drop-In Session Leader in Continuum Mechanics and fluid dynamics of climate and environment.
University of Cambridge, 2023-2024.
Supervisor for undergraduate students in Dynamic and Relativity.
Part III Drop-In Session Leader in Continuum Mechanics.